

void Test(int n) — $T(n)$

```

{
  if (n > 1)
  {
    for (i = 0; i < n; i++)
    {
      stmt; —————  $n$ 
    }
    Test(n/2) —————  $T(n/2)$ 
    Test(n/2) —————  $T(n/2)$ 
  }
}

```

$$T(n) = 2T(n/2) + n$$

Full recurrence relation

$$T(n) = \begin{cases} 1 & , n = 1 \\ 2T(n/2) + n & , n > 1 \end{cases}$$

Substitution Method

$$T(n) = 2T(n/2) + n$$

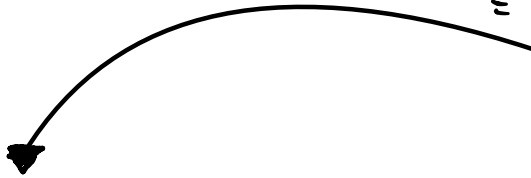
$$? \quad T(n) = 2T(n/2) + n$$

$$T(n/2) = 2T(n/2^2) + n/2$$

$$T(n) = 2[2T(n/2^2) + n/2] + n$$

$$= 2^2 T(n/2^2) + n$$

$$= 2^2 T(n/2^2) + 2n$$



$$= 2^2 T(n/2^2) + 2n$$

$$\stackrel{?}{\rightarrow} T(n/2^2) = 2T(n/2^3) + n/2^2$$

$$= 2^2 [2T(n/2^3) + n/2^2] + 2n$$

$$= 2^3 T(n/2^3) + 3n$$

Assume
this

$$2^k T(n/2^k) + kn$$

$$T(n/2^k) = T(1)$$

$$\frac{n}{2^k} = 1$$

$$n = 2^k$$

$$k = \log n$$

$$T(n) = 2^k T(n/2^k) + kn$$

$$T(n) = 2^k T(1) + kn$$

$$= 2^k \cdot 1 + n \log n$$

$$= n \cdot 1 + n \log n$$

$$= n + n \log n$$

$$O(n \log n)$$

08/05/25 - Video 2.4.1 - Master Theorem in Algorithms for Dividing Functions

General Form of Recurrence Relation

$$\Rightarrow T(n) = aT(n/b) + f(n)$$

$$a \geq 1$$

$$b > 1$$

$$f(n) = \Theta(n^k \log^p n)$$

Now the Master Theorem focuses on the general form and builds on it.

① $\log_b^a$

Case 1: if $\log_b^a > k$, then $\Theta(n \log_b^a)$

② k

Case 2: if $\log_b^a = k$

↳ if $p > -1$, $\Theta(n^k \log^{p+1} n)$

↳ if $p = -1$, $\Theta(n^k \log \log n)$

↳ if $p < -1$, $\Theta(n^k)$

Case 3: if $\log_b^a < k$

↳ if $p \geq 0$, $\Theta(n^k \log^p n)$

if $p < 0$, $\Theta(n^k)$

Example 1: $T(n) = 2T(n/2) + 1$

$$\hookrightarrow a=2, b=2, \underline{f(n)=1}$$

$$f(n) = \underline{\Theta(n^0 \log^0 n)}$$

$k=0, p=0$

now need to find
 $\log_b^a \rightarrow \log_2^2 = 1$

Case 1: $\Theta(n^{\log_b^a})$

$$\Theta(n^1) \rightarrow \boxed{\Theta(n)}$$

we have k and $\log_b^a$
 now which case?

$1 > 0$, so case 1.

Example 2: $T(n) = 4T(n/2) + n$

$$\hookrightarrow a=4, b=2, \underline{f(n)=n}$$

$$\Theta(n^1 \log^0 n)$$

$k=1, p=0$

$$\log_b^a = \log_2^4 = 2$$

so using masters, $2 > 1$. So
 this is case 1.

$$\Theta(n^{\log_b^a})$$

$$= \boxed{\Theta(n^2)}$$

Example 3: $T(n) = 2T(n/2) + n$

$\hookrightarrow a=2, b=2, f(n)=n$

$\Theta(n^1 \log^0 n)$

$\hookrightarrow k=1, p=0$

$\log_b^a = \log_2^2 = 1$

now compare $\log_b^a$ and k ,
they're equal so $1 = 1$.

Case 2

Case 2: $\Theta(n^k \log^{p+1} n) \rightarrow$ this is $p > -1$ since
 $p=0$

$\downarrow$
 $\boxed{\Theta(n \log n)}$

Case 1

$$\hookrightarrow T(n) = 2T(n/2) + 1 \quad \text{---} \quad \Theta(n)$$

$$T(n) = 4T(n/2) + 1 \quad \text{---} \quad \Theta(n^2)$$

$$T(n) = 4T(n/2) + n \quad \text{---} \quad \Theta(n^2)$$

$$T(n) = 8T(n/2) + n^2 \quad \text{---} \quad \Theta(n^3)$$

$$T(n) = 16T(n/2) + n^2 \quad \text{---} \quad \Theta(n^4)$$

Case 3

$$\hookrightarrow T(n) = T(n/2) + n \quad \text{---} \quad \Theta(n)$$

$$T(n) = 2T(n/2) + n^2 \quad \text{---} \quad \Theta(n^2)$$

$$T(n) = 2T(n/2) + n^2 \log n \quad \text{---} \quad \Theta(n^2 \log n)$$

$$T(n) = 4T(n/2) + n^3 \log^2 n \quad \text{---} \quad \Theta(n^3 \log^2 n)$$

$$T(n) = 2T(n/2) + n^2 / \log n \quad \text{---} \quad \Theta(n^2)$$

Case 2

$$T(n) = T(n/2) + 1 \quad \text{—————} \quad \Theta(\log n)$$

$$T(n) = 2T(n/2) + n \quad \text{—————} \quad \Theta(n \log n)$$

$$T(n) = 2T(n/2) + n \log n \quad \text{—————} \quad \Theta(n \log^2 n)$$

$$T(n) = 4T(n/2) + n^2 \quad \text{—————} \quad \Theta(n^2 \log n)$$

$$T(n) = 4T(n/2) + (n \log n)^2 \quad \text{—————} \quad \Theta(n^2 \log^3 n)$$

$$T(n) = 2T(n/2) + n / \log n \quad \text{—————} \quad \Theta(n \log \log n)$$

$$T(n) = 2T(n/2) + n / \log^2 n \quad \text{—————} \quad \Theta(n)$$